

Programming Exercise: Coupon Collector

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Background

There are $m = 100$ distinct trading cards. Each cookie box contains exactly one card, chosen uniformly at random and independently of all other boxes. Let

N = the number of boxes needed to collect all m distinct cards.

The random variable N is known as the *coupon collector time*.

Tasks

1. Analytical pmf

The probability mass function of N can be expressed using inclusion–exclusion:

$$\mathbb{P}(N = k) = \frac{m!}{m^k} S(k - 1, m - 1), \quad k = m, m + 1, \dots$$

where $S(n, r)$ denotes the *Stirling numbers of the second kind* (the number of ways to partition an n -element set into r non-empty subsets).

Implement a function `pmf_analytical(k, m=100)` that returns $\mathbb{P}(N = k)$.

2. Monte Carlo simulation

Write a function `simulate_one_run(m=100)` that simulates a single collection process and returns the number of boxes opened until all m cards have been seen.

Then, run 10^5 independent replications to obtain an empirical distribution of N . Try to use `set` data type.

3. Comparison plot

4. Plot the empirical pmf $\hat{p}(k)$ (e.g., as a stem or bar plot) for k ranging from m up to $k = 800$).

5. Overlay the analytical pmf $p(k)$ on the same axes (use a different marker or line style).

6. Add a legend, axis labels, and a title.

7. Probability estimate

Using both approaches, estimate

$$\mathbb{P}(N \leq 500).$$

8. From simulation: compute the fraction of runs with $N \leq 500$.
9. From the analytical pmf: sum $p(k)$ for $k = m, \dots, 500$. Report both numbers and comment on their agreement (e.g., absolute difference).

10. Expected value check

The theoretical expectation is

$$\mathbb{E}[N] = m H_m = m \sum_{i=1}^m \frac{1}{i},$$

where H_m is the m -th harmonic number. Compute this value and compare it with the sample mean from your simulation.

Tips

1. **Stirling numbers:** In Python, `sympy.functions.combinatorial.numbers.stirling` provides $S(n, r)$. Alternatively, implement the recurrence

$$S(n, r) = r S(n-1, r) + S(n-1, r-1), \quad S(0, 0) = 1, \quad S(n, 0) = S(0, r) = 0 \quad (n, r > 0).$$

2. **Numerical stability:** For large k , directly evaluating $\frac{m!}{m^k} S(k-1, m-1)$ may underflow. Work in log-space if needed.